SIH26143 FINAL IMPLEMENTATION VALIDATION

Maritime Oil-Spill Attribution System — Research → Engineering Freeze

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Executive Verdict

The core pipeline (SAR segmentation → drift reconstruction → AIS correlation → candidate ranking) is **not novel as a concept** — it is the same approach used operationally by EMSA CleanSeaNet since 2007 and SMHI's Seatrack Web, and published repeatedly in the marine-science literature (Seatrack Web backtracking + AIS, 2008; bidirectional drift-similarity ship tracing, 2024). This is good news, not bad news: it means your architecture is scientifically defensible, not speculative. **RESEARCH FINDING**, cited below.

What is *not* already a solved, published, off-the-shelf system is: (1) doing this with **quantified, Monte-Carlo uncertainty on the attribution itself** rather than a single deterministic backtrack, and (2) an **explainable, evidence-for/evidence-against scoring layer** with an explicit AIS-gap-is-not-guilt safeguard, packaged as a coherent forensic investigation tool rather than a research prototype. That combination is a legitimate, defensible SIH differentiator — but it is an **engineering and rigor differentiator**, not a scientific breakthrough. Say that honestly to evaluators. Do not claim to have invented drift-based polluter attribution; claim to have made it uncertainty-aware, explainable, and safe to use as evidence rather than accusation.

DECISION: Build the full pipeline, but the innovation pitch is "uncertainty-quantified, explainable attribution" — not "AI detects oil spill polluters" (this already exists operationally in Europe).

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1. Attribution Scoring

Research Findings

Published bidirectional-drift ship-tracing work (Yang et al. 2024, *Marine Pollution Bulletin*) computes forward-drift similarity between candidate vessel trajectories and the observed spill footprint, rather than hand-picked weights. Seatrack Web / EMSA CleanSeaNet (SMHI, since 2007) correlate hindcast/forecast drift with AIS positions and use analyst review for final determination — they do **not** fully automate the guilt decision. **RESEARCH FINDING.**

Recommended Method

Two-stage: **(1)** a small set of physically-grounded features feeding **(2)** a transparent, additive scoring function, not a black-box classifier. A six-person team cannot responsibly validate a learned ranker (logistic regression / learning-to-rank) with the tiny, label-scarce dataset this domain has — there is no ground-truth "confirmed polluter" training set publicly available. **ENGINEERING RECOMMENDATION.**

A. Features to include (all measurable, all defensible):

- Spatial proximity: minimum distance between vessel's simulated/backward position and reconstructed spill origin at estimated spill time
- Temporal proximity: time offset between vessel's presence at that location and estimated spill age
- Trajectory–drift compatibility: distance between vessel's forward-simulated slick (using the same environmental forcing) and the observed spill polygon (this is the bidirectional-similarity idea from Yang et al. — most defensible single feature)
- Speed/behavior anomaly: deceleration, loitering, or course reversal near the estimated origin
- AIS continuity/gap score (module 3, injected here as one feature, capped in weight — see safety limitation)
- Vessel type/cargo plausibility (tanker/bulk carrier weighted higher than passenger vessel — used only as a plausibility multiplier, never a primary score)

B. Features to exclude:

- Vessel size/DWT as a direct score input — too weak a signal, correlates with normal traffic composition, easily biases against large legitimate tankers
- Nationality/flag state — not scientifically defensible, legally and ethically hazardous, drop entirely
- Past violation history — useful for a human investigator's judgment, dangerous as an automated score input (prior-bias / presumption of guilt); expose it only as separate context, never fused into the number

C. Score construction — recommended: weighted-linear MVP, Bayesian upgrade for SIH-final. Rule-based alone is too brittle (fails silently outside its rule set) and full ML is unvalidatable at this data scale. A **weighted, normalized, bounded scoring function** for MVP, escalating to a **naive-Bayes / Bayesian updating** approach for the SIH-final round, is the correct middle path.

D. Weight derivation — do not invent numbers.

- Do **not** hand-pick weights and call it done.
- MVP: derive weights via **AHP (Analytic Hierarchy Process) pairwise comparison** among your own team + one domain expert (INCOIS/Coast Guard contact if reachable) — this is a documented, defensible, citable method for turning expert judgment into weights, and is standard practice in environmental-risk scoring.
- SIH-final: run a **sensitivity analysis** (vary each weight ±20% and show ranking stability) and report it — evaluators respect "we tested whether our weights matter" far more than a fixed number with no justification.
- Advanced: if you build ≥2 synthetic scenarios (see Point 4), fit weights via **logistic regression on synthetic labeled cases** and report it explicitly as "trained on synthetic data — not operationally validated," which is honest and still impressive.

E. Recommended MVP scoring formula:

```
Score(v) = w1·Proximity_norm + w2·TemporalMatch_norm + w3·DriftCompatibility_norm
           + w4·BehaviorAnomaly_norm + w5·(1 − AISGapPenalty)
subject to: w1+w2+w3+w4+w5 = 1, all components normalized to [0,1]
Suggested AHP-derived starting weights: w3=0.30 (drift compatibility is the strongest physical signal),
w1=0.20, w2=0.20, w4=0.15, w5=0.15
```

Vessel-type plausibility applied as a bounded multiplier (0.85–1.0), never additive, so it can only slightly down-weight implausible candidates, never manufacture a positive score from nothing.

F. Advanced SIH approach: Bayesian belief updating — start from a uniform prior over all vessels in the traffic lane at spill time, update sequentially on each evidence feature as an independent (or lightly correlated) likelihood ratio, output posterior probability per vessel. This directly composes with the Monte-Carlo layer (Point 2): each simulation run produces one posterior ranking; aggregate across runs for stability.

G. Explainability — mandatory, not optional. Every candidate must ship a feature-attribution breakdown. Example output:

```
Vessel: MV [REDACTED], MMSI xxxxxxxxx
Attribution Score: 0.71 (Rank 1 of 4 candidates)
Why:
  + Drift compatibility: 0.34/0.30 weight — forward-simulated slick overlaps
    observed polygon by 68%
  + Temporal proximity: within 40 min of estimated spill age
  + Speed anomaly: 3.1 kt deceleration observed near estimated origin
  − AIS continuity: no gap detected (penalty = 0, neutral)
  ~ Vessel type: tanker (plausibility multiplier 1.0, neutral)
Confidence caveat: spill age itself is estimated ±2h — see Monte-Carlo section.
```

Validation

Score outputs must be checked against the golden demo scenario (Point 4) — if the known/assumed "true" candidate does not rank first there, the formula is wrong, not the scenario.

DECISION: Weighted-linear (AHP-derived, sensitivity-tested) scoring for MVP; Bayesian posterior updating, composed with Monte-Carlo, for SIH-final. No unvalidated learned ranker.

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2. Monte-Carlo Attribution Stability

Research Findings

OpenDrift/OpenOil (MET Norway) is built specifically for ensemble/uncertainty-aware trajectory modeling — it natively supports seeding elements with a positional uncertainty radius and time window, and is designed to be forced by ensemble ocean/wind products such as CMEMS ensemble reanalysis. Independent validation work comparing OpenDrift ensemble-forced runs against 41 Sentinel-1 observed slicks (Frontiers, 2023) found ensemble forcing captures uncertainty meaningfully better than a single deterministic run. Separately, oil-spill-modeling review literature reports realistic empirical ranges for oil physical/weathering parameters (e.g., oil density roughly 0.2–2.5, kinematic viscosity 5–200 cP, evaporation-relevant temperature ranges) that should ground your perturbation bounds rather than invented percentages. **RESEARCH FINDING**, cited above.

Recommended Method

Perturb the physically uncertain **inputs**, not the algorithm. Run the same OpenOil physics N times with resampled forcing/initial conditions; do not build a custom Monte-Carlo engine.

Parameters

Perturb, with justification:

- **Spill age / detection-to-origin time lag** — dominant uncertainty; SAR revisit gaps mean the spill could be hours to over a day old at detection. Sample from a bounded distribution informed by the actual revisit interval of your chosen SAR pass (this is your single most important uncertain variable — treat it as such).
- **Wind speed/direction** — sample from the spread of the ECMWF/GFS ensemble members (or ERA5 uncertainty estimate) valid at spill time, not an invented percentage.
- **Ocean current speed/direction** — sample from CMEMS ensemble product spread where available; if only a single deterministic CMEMS field is accessible for your MVP, apply a documented perturbation (e.g., ±20% magnitude, ±15° direction) explicitly labeled as an **engineering approximation**, not a scientific fact, until ensemble current data is integrated.
- **Initial spill polygon centroid** — perturb within the SAR pixel/geolocation uncertainty (tens to a few hundred meters, per sensor spec) plus segmentation-model boundary uncertainty.
- Do **not** perturb oil weathering parameters heavily for MVP — they matter for spill *age estimation*, not for backward drift direction, and add complexity your six-person team cannot validate in the time available.

Simulation Strategy

- **MVP / demo: 200–500 runs.** Compare: 100 is noisy for a stability percentage (coarse buckets like 82% vs 87% aren't meaningfully different); 5000 is unnecessary given only ~4-5 perturbed input distributions and adds runtime/hardware burden with no accuracy gain at this scenario complexity; 500 is the practical sweet spot for laptop-class hardware within a live demo's time budget.
- **SIH-final: 1000 runs offline, pre-computed for the golden scenario**, with the live demo replaying/visualizing the precomputed ensemble rather than recomputing 1000 physics runs on stage — this is a legitimate and common practice (avoids demo-day compute risk) and should be stated openly as such.

Stability Formula

Your proposed ratio (simulations where vessel X is a valid candidate ÷ total valid simulations) is directionally right but **incomplete on its own** — it treats "top candidate" as binary per run and discards margin information. Recommended refinement:

```
StabilityScore(v) = (1/N) · Σ_i  [rank_i(v) == 1]          → "win rate" (primary, reportable %)
MarginScore(v)    = (1/N) · Σ_i  (Score_i(v) − Score_i(runner-up))   → confidence-of-win magnitude
Report both. A vessel with 90% win rate but near-zero margin is a much weaker
claim than 90% win rate with a large, consistent margin.
```

"Hit" Definition

A candidate counts as a valid hit in a simulation run if: spatial distance between simulated position and spill polygon centroid < threshold (start with SAR pixel resolution × 3, tune later) **AND** temporal offset within the run's sampled spill-age window **AND** a minimum drift-compatibility overlap (e.g., >30% polygon overlap) is met. Require all three jointly — any single criterion alone produces too many false hits in busy shipping lanes.

Edge Cases

- **Multiple high-scoring vessels:** report all above a stability threshold (e.g., >50% win rate) as "co-candidates," do not force a single name.
- **No matching vessels:** output `INSUFFICIENT EVIDENCE TO ATTRIBUTE` explicitly, with the reconstructed drift cone shown for context — this is a required output state, not a failure mode.
- **Incomplete AIS:** widen the candidate search radius/time window and flag reduced confidence explicitly in the report.
- **Uncertain spill age:** this is already the dominant perturbed variable — its uncertainty is absorbed into the Monte-Carlo run, not treated as a separate failure.
- **Poor weather data:** fall back to climatological wind/current averages for the region/season, flag the run as "low-confidence forcing" in metadata.
- **Model failure (e.g., OpenDrift crash on a run):** discard that run, do not count it in N; log failure rate — a high failure rate itself is diagnostic and should be surfaced.

Is "Attribution Stability" genuinely novel?

Partially. Ensemble/uncertainty-aware drift forecasting exists and is published (Frontiers 2023, OpenDrift literature). What is not commonly packaged is **turning ensemble drift uncertainty into a per-vessel stability/confidence score that gates whether attribution is even claimed.** That packaging — refusing to name a polluter when evidence is unstable — is your genuine, honestly-stated differentiator. Do not oversell it as inventing Monte-Carlo drift modeling; that already exists.

DECISION: 500-run live/demo ensemble, 1000-run precomputed for SIH-final golden scenario; perturb spill age, wind, current, initial position; report win-rate AND margin, not win-rate alone; explicit `INSUFFICIENT EVIDENCE` output state is mandatory.

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3. AIS Gap / Dark Vessel Analysis

Research Findings

The xView3-SAR dataset and follow-on work (NeurIPS 2022; "DarkVesselNet," 2026) explicitly frame AIS absence as an **evidence-fusion-under-missingness problem**, not a binary guilt signal — AIS gaps are driven by coverage limitations, receiver density, device failure, weather, legal non-carriage, and deliberate disabling, all mixed together, and the correct model treats missingness as context-dependent rather than a single anomaly flag. Global Fishing Watch's operational "dark vessel" work independently confirms that a large share of vessels are AIS-dark for entirely mundane reasons (coverage, non-carriage requirements) at any given time, especially outside dense shipping lanes. **VERIFIED FACT / RESEARCH FINDING**, cited above.

Recommended Method

Build a 3-way classifier for every observed gap, **not** a 2-way (gap/no-gap) flag:

- **NORMAL GAP:** duration and location consistent with known coverage limitations (e.g., far from AIS base-station range, satellite revisit gap, documented small-vessel non-carriage threshold)
- **SUSPICIOUS GAP:** gap begins/ends near the estimated spill time and location, with a course/speed change bracketing the gap, in an area with otherwise good AIS coverage
- **UNKNOWN GAP:** insufficient context to classify — the honest default when signals conflict

Safe Features

Gap duration; distance from spill origin at gap start/end; timing offset relative to estimated spill age; speed/heading continuity immediately before vs. after the gap (large discontinuity is more suspicious than smooth continuity); local AIS receiver/coverage density in that area (if available, even coarsely); whether a SAR pass independently detected an unmatched vessel-like object at that location during the gap (see below).

SAR–AIS Fusion for Unmatched Vessels

Yes — this is exactly what xView3-SAR operationalized: SAR vessel detection (even simple CFAR/backscatter-threshold detection is enough for MVP; a lightweight detector is sufficient — full xView3-scale deep detection is out of scope for six students) combined with AIS-to-SAR matching. Practical pipeline: (1) run vessel detection on the same/nearby-time SAR scene used for spill detection, (2) project all AIS-reporting vessels' expected positions onto the scene, (3) any detected SAR object with no matching AIS track within a tolerance radius is an "unmatched detection" — feed its presence/absence as one input to the AIS-gap score, not as a standalone accusation.

AIS-Gap Score

```
GapScore(v) = f(duration_norm, proximity_to_spill_norm, timing_alignment_norm,
                behavior_discontinuity_norm, unmatched_SAR_detection_boost)
Output bounded [0,1], capped contribution to final attribution score (see below).
```

Effect on Final Attribution Score

Cap its weight explicitly (recommended ≤15% of total score, per the MVP formula in Point 1) and never let it single-handedly produce a top rank — it must always co-occur with at least one physical drift/proximity feature to matter. This is a hard architectural constraint, not a tuning suggestion.

What the System MUST NOT Claim

- It must never output or imply "AIS gap = guilty" or "AIS gap = dumping."
- Correct phrasing, hard-coded into report templates: *"An AIS reporting gap was observed for this vessel during the relevant window. This is one contributing factor among several and does not, on its own, establish responsibility. Gaps commonly occur for non-suspicious operational reasons."*
- The system must never present a single-vessel accusation as a finding without the accompanying stability score and explanation.

DECISION: 3-way gap classification (normal/suspicious/unknown); AIS-gap score capped at ≤15% of total attribution weight; hard-coded non-accusatory language in every report; SAR-unmatched-detection as a bonus input, not a standalone flag.

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4. Dataset & Golden Demo Scenario

Dataset Comparison Table

Dataset	Source	Coverage	Free?	MVP-suitable?	Final-demo?	Limitations
Krestenitis oil-spill SAR benchmark	MKLab/CERTH (Remote Sensing, 2019)	European waters, 1,112 Sentinel-1 scenes, 5-class masks	Free on request (institutional email)	Yes — request access immediately, lead time exists	Yes, as training data	Not India-specific; request approval delay
Deep-SAR Oil Spill	Public research	ALOS PALSAR + Sentinel-1A, 8,070	Free	Yes	Yes	Small crop size may need re-mosaicking

(SOS) dataset	release	cropped 256×256 images			(supplementary)	for realistic scene-level demo
Sentinel-1 SAR Oil Spill dataset (Zenodo)	Zenodo	1,200 train/val Sigma0 GeoTIFF pairs	Free, direct download	Yes	Yes	Regional bias (non-Indian waters), same as above
ESA Copernicus Open Access Hub / Copernicus Data Space	ESA	Global, incl. Indian Ocean, since Oct 2014	Free, registration only	Yes — for pulling your own raw Indian-water Sentinel-1 scenes	Yes	You must find/label your own spill or use a known past incident scene; no pre-labeled Indian spill masks exist publicly
ERA5 reanalysis	ECMWF/Copernicus Climate Data Store	Global, hourly, 1940–present	Free, CDS API (registration)	Yes	Yes	API quota/queue delays — download early, don't leave for demo week
Copernicus Marine Service (CMEMS) ocean currents	Copernicus Marine Service	Global + Indian Ocean regional products	Free, registration	Yes	Yes	Ensemble/uncertainty products less complete for Indian Ocean than for European seas
Global Fishing Watch AIS Vessel Presence / apparent fishing effort	Global Fishing Watch	Global, 2012–2024, hourly-binned	Free (API/Zenodo, some products request-based)	Yes — global context, dark-vessel base rates	Yes	Aggregated/gridded, not full raw point-track granularity everywhere; MMSI-vessel identity linkage has known limitations (flag-state ambiguity)
Danish Maritime Authority historical AIS	DMA	Danish waters, full raw point tracks since 2014	Free, direct daily CSV download	Yes — for pipeline development/testing only	No (wrong ocean)	Not Indian waters at all — use purely to build and test your AIS ingestion/analytics code before wiring in real target-region data
INCOIS Oil Spill Advisory / SARAT	ESSO-INCOIS (MoES)	Indian Ocean, operational forecast service	Advisory service, not an open bulk-download dataset	No (not a downloadable dataset)	No	It's a live advisory *service* for the Coast Guard/Navy, not a public archive you can bulk-download — useful for context/legitimacy in your pitch, not as raw data

Recommended Dataset Stack

Sentinel-1 GRD scenes (own-pulled via Copernicus Data Space, targeting Indian coastal waters) for the demo scene; Krestenitis + SOS + Zenodo datasets pooled for **training/fine-tuning your segmentation model** (none of them are Indian-water, and that

is fine — segmentation models trained on European SAR spill signatures transfer reasonably to other oceans because the physics of oil-on-water backscatter is not region-specific, even though generalization gaps are a documented open research issue); ERA5 for wind; CMEMS for currents; Danish AIS data to build/test your ingestion pipeline; then substitute synthetic AIS for the actual Indian-water demo (see below).

Indian-Water Strategy

RESEARCH FINDING, verified: there is no public, pre-packaged Indian-water dataset that pairs a labeled oil-spill SAR scene with a real, complete, downloadable AIS archive and a confirmed "true" polluter — INCOIS provides an operational advisory *service*, not a public forensic dataset, and no equivalent of EMSA CleanSeaNet's internal archive is publicly downloadable for Indian waters.

Comparing your four options:

- **Option A (real Indian historical case):** Not feasible for a student team in this timeframe — you cannot obtain confirmed ground truth or a full AIS archive for a real past Indian incident without institutional access (Coast Guard/INCOIS cooperation), and fabricating certainty about a real, named incident where you cannot verify the actual responsible party is both scientifically dishonest and potentially defamatory. **Do not do this.**
- **Option B (real international historical case):** Scientifically honest and immediately usable — e.g., a known, published tanker-source spill case with public AIS and SAR imagery available (such cases exist in the Baltic/Mediterranean literature already cited above, where the "true" vessel was confirmed by authorities). Good for **validating your methodology**, weak for the "India-oriented" pitch.
- **Option C (fully synthetic scenario, built on real Indian environmental data):** Best fit for your actual constraint. Use a real Sentinel-1 scene over Indian waters (with or without a real minor slick/look-alike present — even a look-alike scene is fine if clearly labeled synthetic), real ERA5/CMEMS forcing for that date/location, and a **synthetic AIS scenario you construct yourself** with clearly documented, plausible vessel tracks. This is scientifically honest as long as you **label it as synthetic everywhere, including in the live demo narration.**
- **Option D (hybrid: real scene + real environment + synthetic AIS + one validated real international case as a methodology cross-check): This is the correct answer.**

DECISION:

- **MVP:** Option C (synthetic AIS on real Indian environmental/SAR data).
- **University demo / SIH final:** Option D — lead with the Option-B real international case to prove the pipeline is scientifically valid against a known answer, then present the Option-C Indian-water synthetic scenario as "how this would run operationally in Indian waters," explicitly labeled synthetic throughout.
- **Research validation:** Option B is your validation case — report accuracy against it honestly, including failure modes.
- **Never claim a real, unverified Indian incident has a resolved "guilty vessel."**

Golden Demo Scenario

Define this in your first team meeting and freeze it — do not let it drift as different members build different parts:

```
SAR image:         One real Sentinel-1 GRD scene, Indian coastal waters
                    (e.g., Gulf of Kutch, Mumbai High offshore field approach,
                    or Chennai/Ennore approach — pick based on actual traffic
                    density visible in AIS Vessel Presence data)
Spill location:    A synthetically injected spill polygon (or a real look-alike
                    feature, clearly relabeled) at a chosen coordinate
Timestamp:         Matches the real SAR acquisition time
Spill age:         Assumed 6–18h unknown, sampled in Monte-Carlo per Point 2
Environmental data: Real ERA5 wind + real CMEMS currents for that exact date/location
AIS dataset:       Synthetic — 8–12 constructed vessel tracks, 1 designed as the
                    "true" source (consistent with drift), 2–3 plausible
                    distractors, 1 with a fabricated AIS gap, rest clean traffic
Candidate vessels: The synthetic fleet above
Expected output:   Top candidate correctly identified with a stated stability
                    score; at least one distractor scores non-trivially
```

```
                  (to show the system isn't trivially rigged); the gap vessel
                  must NOT auto-win purely on its gap
Ground truth /
validation method: Internal — you know which synthetic vessel you designed as
                  "true," so correctness is self-validated by construction;
                  report this honestly as "synthetic ground truth," and
                  separately show the real Option-B case as external validation
```

DECISION: Freeze this exact scenario before any team member writes pipeline code. Changing the demo scenario after backend work starts is the single most common cause of SIH team schedule collapse.

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5. Final Engineering Specification

Architecture (Components)

- **AI/CV module** — SAR preprocessing + segmentation (U-Net or similar; do not build a custom architecture from scratch — fine-tune an existing lightweight U-Net)
- **Geospatial/Physics module** — OpenDrift/OpenOil wrapper, forcing-data ingestion, Monte-Carlo orchestration
- **AIS Analytics module** — ingestion, cleaning, gap detection/classification, trajectory features
- **Attribution Engine** — feature fusion, scoring, explainability output
- **Backend/API** — orchestrates the above, stateless where possible
- **Database** — PostgreSQL/PostGIS
- **Frontend** — map-centric investigation UI
- **Report Generator** — templated PDF/HTML forensic report

Data Flow

```
SAR GeoTIFF → segmentation → spill polygon + confidence
   → (+ spill-age estimate, uncertain) → OpenOil Monte-Carlo backward ensemble
   → per-run drift cones/particle clouds
   → AIS query (candidate vessels in traffic lane at relevant times)
   → per-vessel per-run feature extraction (proximity, drift-compatibility,
      temporal match, behavior, AIS-gap score)
   → per-run ranking → aggregate stability score across N runs
   → explainability payload per candidate
   → investigation visualization + report generation
```

Module Contracts

SAR Segmentation INPUT: Sentinel-1 GRD GeoTIFF · OUTPUT: GeoJSON spill polygon + confidence · DEPENDENCIES: trained U-Net weights · FAILURE: no-detection → explicit “no spill found” state, do not force a polygon.

Drift Simulation INPUT: spill polygon, spill-age distribution, wind/current fields · OUTPUT: N sets of backward particle-cloud trajectories (netCDF/GeoJSON) · DEPENDENCIES: OpenOil, ERA5, CMEMS · FAILURE: forcing-data gap → fall back to climatology, flag run as low-confidence.

AIS Analytics INPUT: raw/synthetic AIS stream for region+time window · OUTPUT: cleaned per-vessel trajectories + gap classification labels · DEPENDENCIES: PostGIS · FAILURE: missing vessel static data (name/type) → proceed with MMSI-only identity, flag incomplete.

Attribution Engine INPUT: per-run drift clouds + candidate trajectories + gap scores · OUTPUT: per-vessel stability score, margin score, explainability breakdown, or `INSUFFICIENT EVIDENCE` · DEPENDENCIES: all above · FAILURE: zero candidates in radius → explicit insufficient-evidence output, never an empty error.

Report Generator INPUT: attribution engine output + visualization snapshots · OUTPUT: PDF/HTML forensic report with hard-

coded non-accusatory template language · FAILURE: missing visualization asset → generate report with placeholder note, never silently omit the uncertainty section.

Database (PostgreSQL/PostGIS) — keep minimal

`incident` · `spill_geometry` (PostGIS polygon) · `spill_metadata` (age assumption, confidence) · `ais_points` · `ais_trajectories` (derived) · `environmental_snapshots` (wind/current refs, not raw grids — store paths/references to netCDF, not the grids themselves, in Postgres) · `candidate_vessels` · `simulation_runs` (one row per Monte-Carlo run, for auditability) · `attribution_scores` · `evidence_items` (per-feature breakdown) · `reports` . **DECISION: Do not store raw environmental grids in PostGIS — reference files on disk/object storage. Overengineering the DB is a top-15 risk (see below).**

API (MVP minimum)

```
POST /incident
POST /spill/segment
POST /drift/simulate        (renamed from "backward" — it runs the full ensemble, not one direction)
POST /ais/query
POST /attribution/run
GET  /investigation/{id}
POST /report/generate
```

Your original list is validated with one addition (`/drift/simulate` replacing `/drift/backward` , since the module must support the forward bidirectional-similarity feature from Point 1 as well as pure backtracking).

Frontend Workflow

1. Select/create incident (choose or upload SAR scene)
2. Trigger processing (segmentation → drift → AIS correlation, shown as a progress pipeline, not a black box spinner)
3. View spill polygon + confidence on map
4. View drift ensemble (cone/heatmap of particle spread, not a single line — this visually communicates uncertainty from the start)
5. View ranked candidates with stability + margin scores
6. Inspect per-candidate evidence breakdown (Point 1's explainability output)
7. View attribution stability panel (win-rate histogram across runs)
8. Generate report (with mandatory non-accusatory language, uneditable in the template's caveat section)

MVP vs Advanced

- **MVP — MUST WORK:** segmentation on a pre-selected scene; single-scenario Monte-Carlo (200–500 runs) with 3–4 perturbed variables; weighted-linear scoring with AHP weights; 3-way AIS gap classification; golden demo scenario end-to-end; PDF report.
- **SIH FINAL — SHOULD WORK:** Bayesian scoring upgrade; 1000-run precomputed ensemble; SAR-unmatched-vessel detection fusion; sensitivity analysis visualization; real-international-case validation panel.
- **FUTURE — DO NOT BUILD NOW:** learned/ML ranking model; live real-time Sentinel-1 tasking integration; multi-incident cross-correlation; mobile app; user auth/multi-tenant system; any claim of legal evidentiary standing.

Top 15 Risks

#	Category	Risk	Prob.	Impact	Mitigation	Fallback
1	Scientific	Overclaiming novelty (drift+AIS attribution already	High	High (credibility	Pitch uncertainty-quantification + explainability as the	Reframe pitch mid-prep if evaluators

		exists operationally)		loss with evaluators)	innovation, not the base pipeline	push back
2	Data	No real Indian ground-truth incident dataset exists	Certain	Medium	Hybrid demo strategy (Point 4)	Lean harder on real international validation case
3	AI/ML	Segmentation model trained on non-Indian SAR generalizes poorly to your target scene	Medium	Medium	Visually validate on your actual demo scene early, not at the end	Manually refine/relabel the polygon for the golden scenario if model underperforms
4	AIS	Synthetic AIS scenario looks obviously fake/rigged to evaluators	Medium	Medium	Include genuine distractor vessels that score non-trivially	Be transparent that it's synthetic — honesty scores better than a thin disguise
5	Physics	OpenDrift/CMEMS ensemble data incomplete for Indian Ocean	Medium	Medium	Document engineering-approximation perturbations (Point 2) explicitly	Use climatological variance if ensemble products unavailable
6	Backend	Monte-Carlo runtime too slow for live demo	Medium	High	Precompute golden-scenario ensemble ahead of time	Live-run only a small illustrative N during demo, show precomputed full results
7	Frontend	Map/drift-cone visualization too complex to build well in time	Medium	Medium	Use existing mapping library (deck.gl/Leaflet + heatmap layer), don't build custom rendering	Static pre-rendered visualization frames if live rendering fails
8	Integration	Six students, five modules — interface mismatches at merge time	High	High	Freeze API contracts (Point 5) before coding starts	Daily integration checkpoint, not a big-bang merge at the end
9	Security	None specifically required for MVP, but demo credentials/API keys exposed in repo	Low	Medium	.env + secrets excluded from GitHub	Rotate any exposed key immediately
10	Performance	Large SAR GeoTIFFs slow to process on laptop-class hardware	Medium	Medium	Downsample/tile for demo scene	Pre-process golden scenario once, cache output
11	Demo	Live network dependency (API calls to CMEMS/ERA5) fails on stage	Medium	High	Cache all golden-scenario data locally before demo day	Fully offline demo mode, no live external calls
12	SIH evaluation	Evaluator asks "how do you know your synthetic scenario's answer is right" and team is unprepared	High	High	Rehearse the honest answer: ground truth by construction + real-case validation	Have the real international validation case ready to show immediately
		Attribution presented as			Hard-coded caveat	Team lead reviews

13	Scientific	legally conclusive rather than investigative-lead	Low if disciplined	High	language everywhere (Point 3)	all report templates before freeze
14	Data	AIS/SAR timestamp or coordinate mismatches (projection errors)	Medium	Medium	Standardize on WGS84 + UTC everywhere from day one	Dedicated "data contract" doc, enforced in code review
15	Backend	PostGIS/spatial query performance untested at demo data volume	Low	Low	Test with realistic-sized synthetic AIS volume early	Reduce candidate search radius/time window if slow

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What We Must NOT Build

- A learned/ML-based attribution ranker (no validated training data exists)
- Any feature using flag state, nationality, or violation history as a direct score input
- A claim of legal/evidentiary conclusiveness
- A live, real-time SAR tasking/ingestion system
- A custom Monte-Carlo or drift-physics engine from scratch (use OpenDrift/OpenOil)
- An over-normalized, multi-table database beyond what's listed in Point 5
- A "definitely guilty" output state, ever, under any configuration

What We Must Freeze Before Coding

- The golden demo scenario (exact scene, coordinates, timestamps, synthetic fleet)
- The API contract in Point 5
- The MVP scoring formula and its weights (even if revisited later, freeze a v1 now)
- The AIS-gap 3-way classification scheme and its capped weight in the final score
- The database schema in Point 5
- The non-accusatory report language templates

Final SIH Verdict

1. **Are we ready to implement?** Yes, conditional on freezing the golden scenario and API contract in the next team meeting — the science underneath is sound and well-precedented.
2. **What must be frozen before coding?** See list immediately above.
3. **What remains scientifically uncertain?** Whether your specific weighted formula/thresholds generalize beyond the golden scenario — you cannot fully resolve this with a six-person team in SIH timelines, and you should say so openly rather than overclaim general validity.
4. **What should we NOT build?** See "What We Must NOT Build" above.
5. **What is the single most important innovation?** Refusing to force an accusation — the `INSUFFICIENT EVIDENCE` state and stability/margin reporting, not the base pipeline.
6. **What is the single biggest technical risk?** Team integration across five modules on a tight timeline (risk #8) — mitigate with a frozen API contract and daily integration checks, not heroics at the end.
7. **What is the single biggest dataset risk?** No public Indian-water ground-truth incident exists (risk #2) — solved by the honest hybrid strategy in Point 4, not by pretending otherwise.

8. **What could cause evaluators to reject/downgrade us?** Presenting the synthetic scenario as if it were real, or overclaiming novelty for the base drift+AIS pipeline, which experienced evaluators may already know is operational practice in Europe.
9. **What is our strongest defensible SIH pitch?** "We built an uncertainty-quantified, explainable maritime attribution system that refuses to accuse without stable evidence — turning a deterministic forensic technique already used by EMSA into a transparent, Monte-Carlo-validated decision-support tool suitable for Indian coastal waters, where no equivalent public system exists today."
10. **What would a technical evaluator challenge you on?** "Your synthetic AIS scenario has a ground truth you invented yourself — how is this validated against reality?" Have the real international validation case (Option B) ready as your answer, and be ready to state plainly what remains unvalidated.

Sources

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- ECMWF/Copernicus Climate Data Store (ERA5) and Copernicus Marine Service (CMEMS) official documentation

Distinctions applied throughout: dataset facts and cited findings above are VERIFIED FACT/RESEARCH FINDING; scoring formulas, weight caps, and architecture choices are ENGINEERING RECOMMENDATION; novelty and risk assessments are INFERENCE based on the cited findings.